Swiss Institute of Bioinformatics

INTRODUCTION TO SEQUENCING DATA ANALYSIS

Quality control (QC) + database retrieval

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Learning outcomes

1

Why QC matters

Spot problems in the raw reads before they spread to every downstream result

2

Key QC tools

FastQC, MultiQC, NanoPlot & co. – what each one reports

3

FASTQ & trimming

Read Phred quality scores and remove adapters and low-quality bases

4

Sequence databases

Find, understand and download public data from INSDC and EGA

Why Quality Control?

1 How is the base quality?

Per-base Phred scores along the read – quality usually drops towards the 3′ end

2 What is the read length?

As expected from the run? How does the distribution look after trimming?

3 Adapters or barcodes?

Read-through into the adapter when the insert is shorter than the read; leftover UMIs

4 Overrepresented sequences?

rRNA, primer dimers, PhiX spike-in, PCR duplicates or contamination

Garbage in, garbage out

Alignment, quantification and variant calling all inherit the problems hidden in the raw reads.

QC is cheap – re-analysis is not.

Check every sample

Dedicated software

Built-in — Manufacturer tools

QC during base calling and run monitoring

- **Illumina** SAV / BaseSpace run metrics (% ≥ Q30, cluster density)
- **ONT** MinKNOW live yield and read-length monitoring
- **PacBio** SMRT Link run QC reports

Illumina — Short reads

Commonly used QC software

- **FastQC** per-base quality, GC content, adapter content, duplication
- **MultiQC** aggregates FastQC and 150+ other tools into one report

ONT — Long reads (ONT)

From the basecaller's sequencing summary

- **pycoQC** run-level metrics: yield, read length, quality over time
- **NanoPlot** read length and quality plots from FASTQ, BAM, summary

ONT & PacBio — Any long-read data

Quick numbers and comparisons

- **NanoStat** summary statistics: N50, mean quality, total yield
- **NanoComp** compare several runs or samples side by side

FASTQ format: four lines per read

FASTA + base Quality = FASTQ

```
@A00123:45:HV2KJDSXY:1:1101:1042:1000 1:N:0:ATCACG
GATTACAGCTTGCCATGCAAGTCGGATCCAGTTACGTAGC
+
FFFFFFFFF:FFFF:FFFF:F:FFF,FF:FF:,F:F,,F:
```

1 **Header**
read ID, instrument, run, flow-cell position, index

2 **Sequence**
the base calls: A, C, G, T (N = unknown)

3 **Separator**
a “+”, optionally repeating the header

4 **Quality**
one ASCII character per base = Phred score

Decode (Phred+33): F → 70 − 33 = Q37 : → 58 − 33 = Q25 , → 44 − 33 = Q11

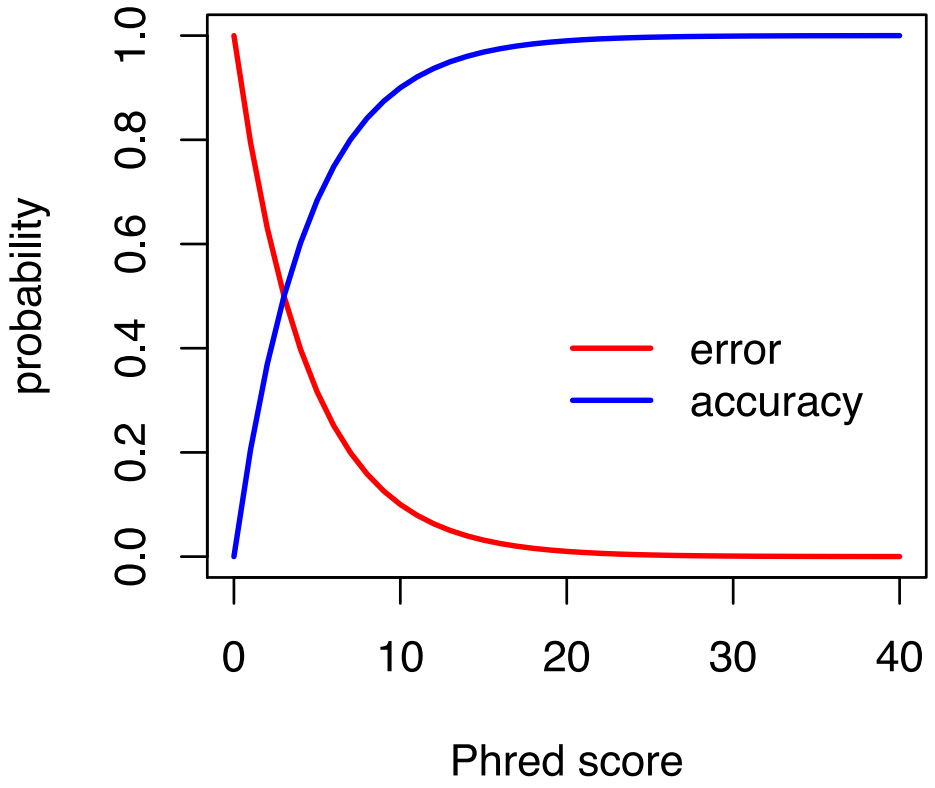
$Q = -10 \cdot \log_{10}(P\ error)$ → Q30 = 1 error in 1,000 bases

FASTQ: Phred quality scores

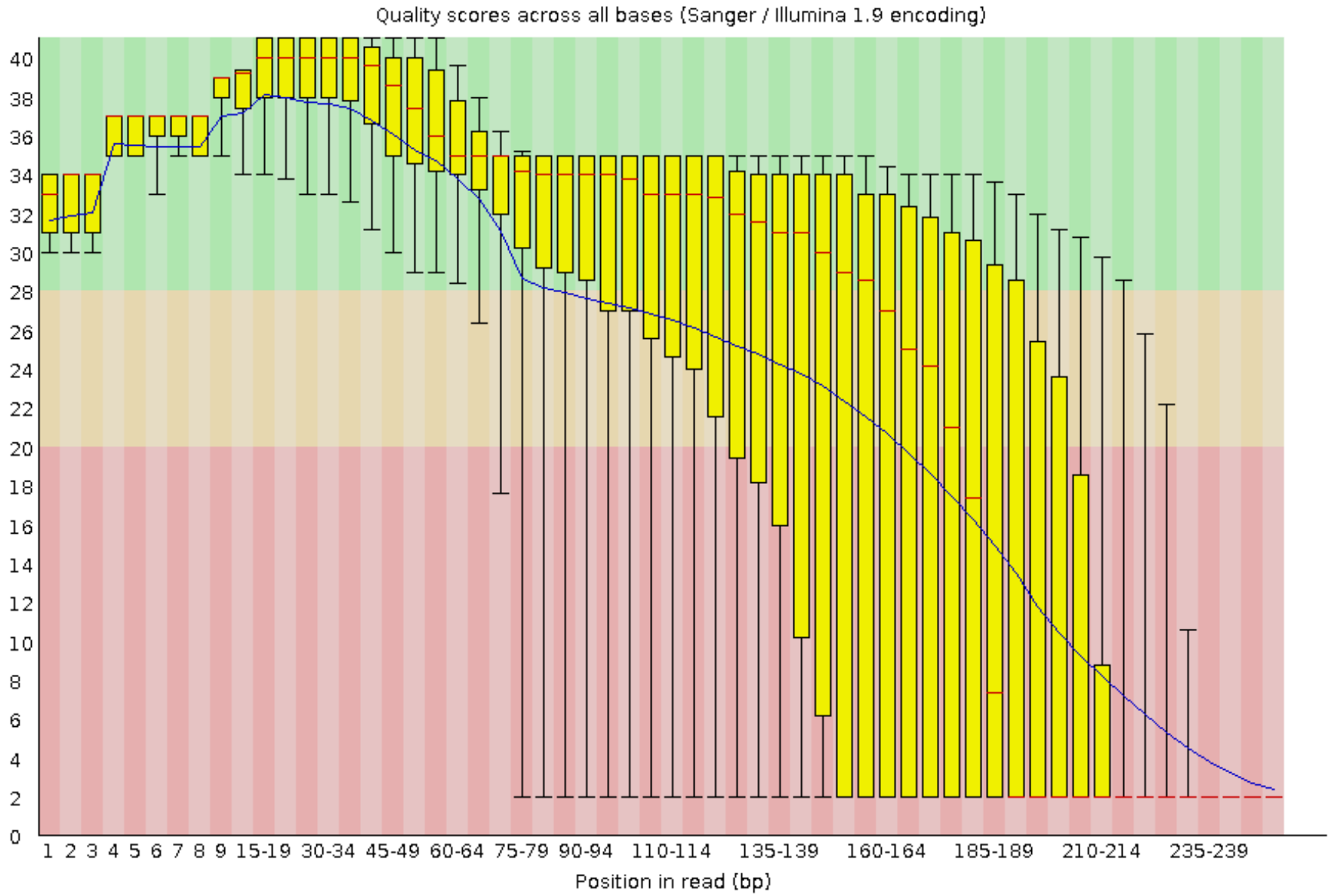
fasta + basequality (fasta + q = fastq)

$$BASEQ = -10log_{10}\Pr\{base\ is\ wrong\}$$

$$-10log_{10}(0.01) = 20$$
$$-10log_{10}(0.1) = 10$$
$$-10log_{10}(0.5) = 3$$



Example of a QC score plot



FastQC zones

Good
Q ≥ 28

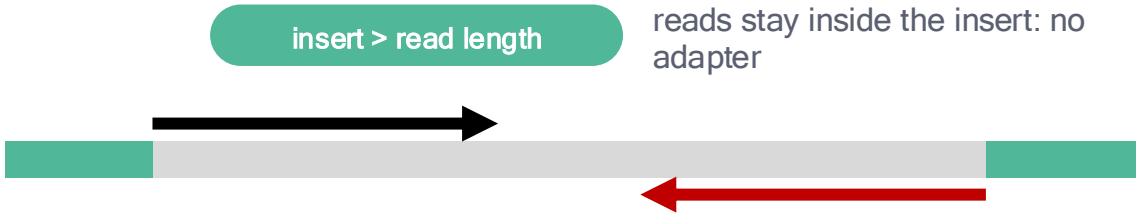
Reasonable
Q 20-28

Poor
Q < 20

Yellow box = 25-75 % range, whiskers = 10-90 %, blue line = mean quality

Adapter sequences

Adapters provide binding sites for primers, allowing for amplification and binding to the flow cell's surface



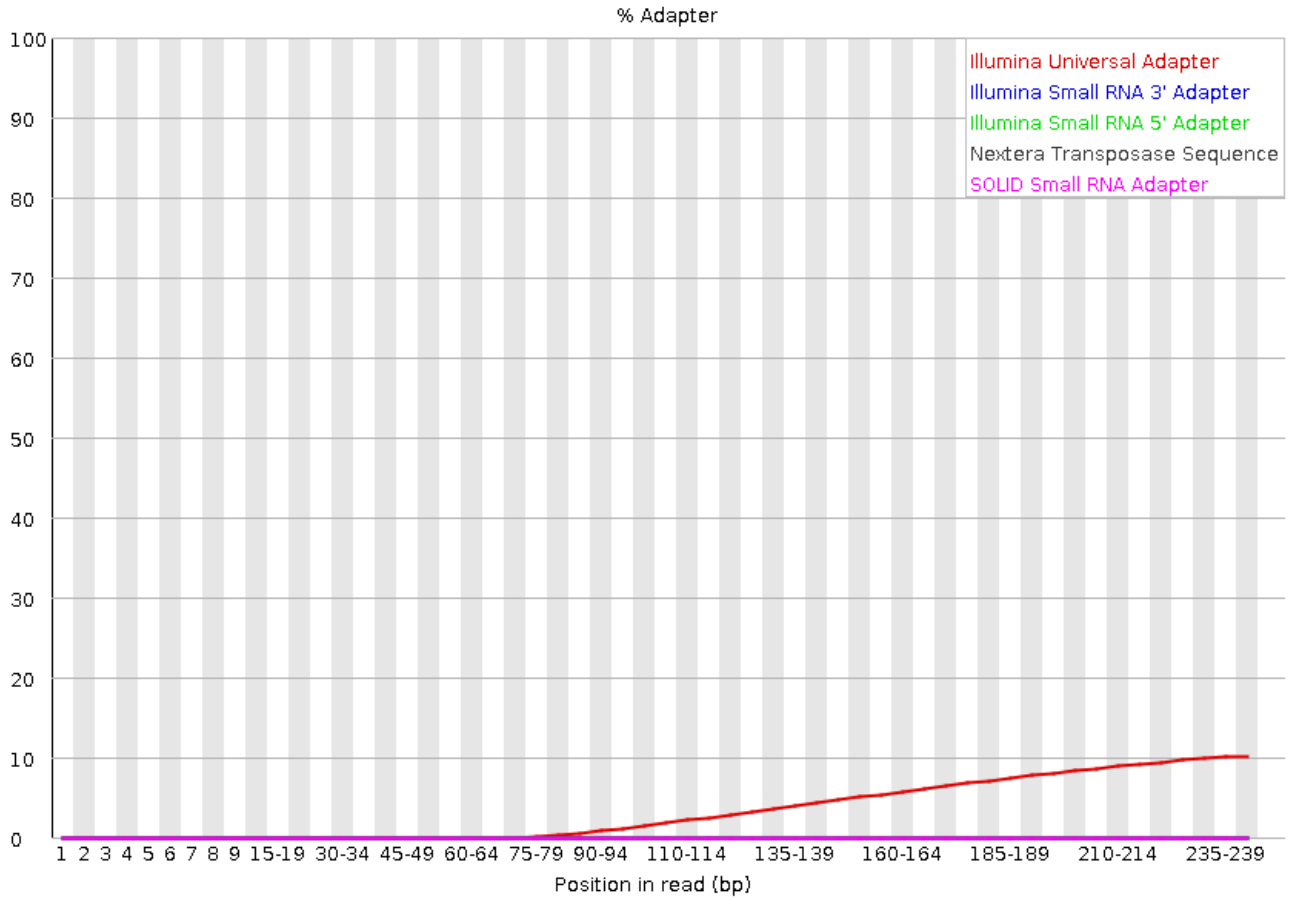
insert > read length

reads stay inside the insert: no adapter



insert < read length

reads run into the opposite adapter (dashed) → adapter content rises in FastQC → trim



Trimming

Find and remove

1 Low base quality

Trim read ends or drop whole reads that fall below a quality threshold

2 Adapter sequences

Read-through into the adapter when the insert is shorter than the read

3 Poly-G tails

2-colour chemistry (NextSeq, NovaSeq) reads “no signal” as G

Software fastp | cutadapt | Trimmomatic | Trim Galore | BBDuk ← *next slide*

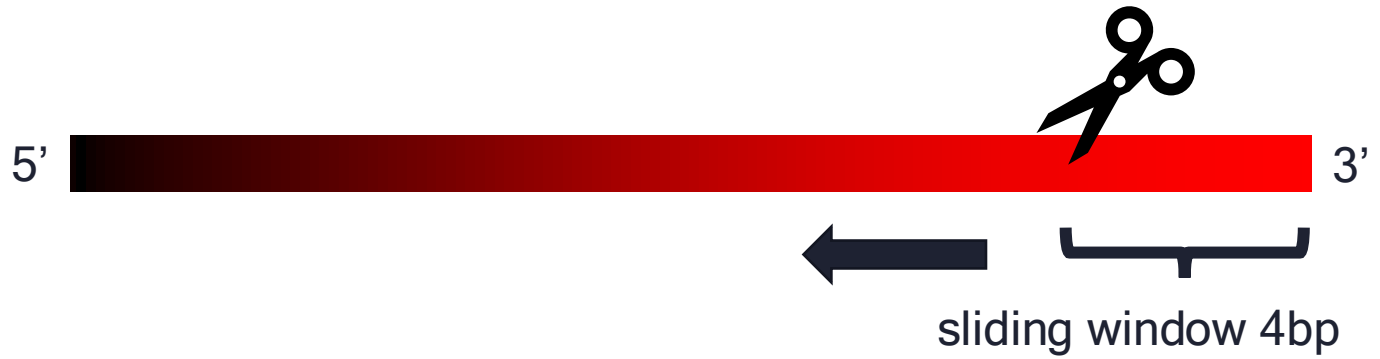
Frequently occurring issues, explained:
sequencing.qcfail.com



Quality trimming with fastp

Default:

- Remove reads with >40% bases <Q15
- Trim poly N (and poly G)
- Autodetect adapters in R1, for both:
  --detect_adapter_for_pe



‘Classical’ trimming: sliding window

- options --cut_front and --cut_tail

Quiz: 13 - 17

What is the primary purpose of sequencing QC?

. . .

Which tool is commonly used for trimming and adapter removal?

Databases



INSDC: International Nucleotide Sequence Database Collaboration

What is INSDC?

International Nucleotide Sequence Database Collaboration

Free and unrestricted access to all public nucleotide sequence data, worldwide

Japan · Asia

DDBJ

National Institute of Genetics, Mishima

Raw reads: **DRA**



Europe

ENA

EMBL-EBI, Hinxton (UK)

Raw reads: **ENA**



USA · North America

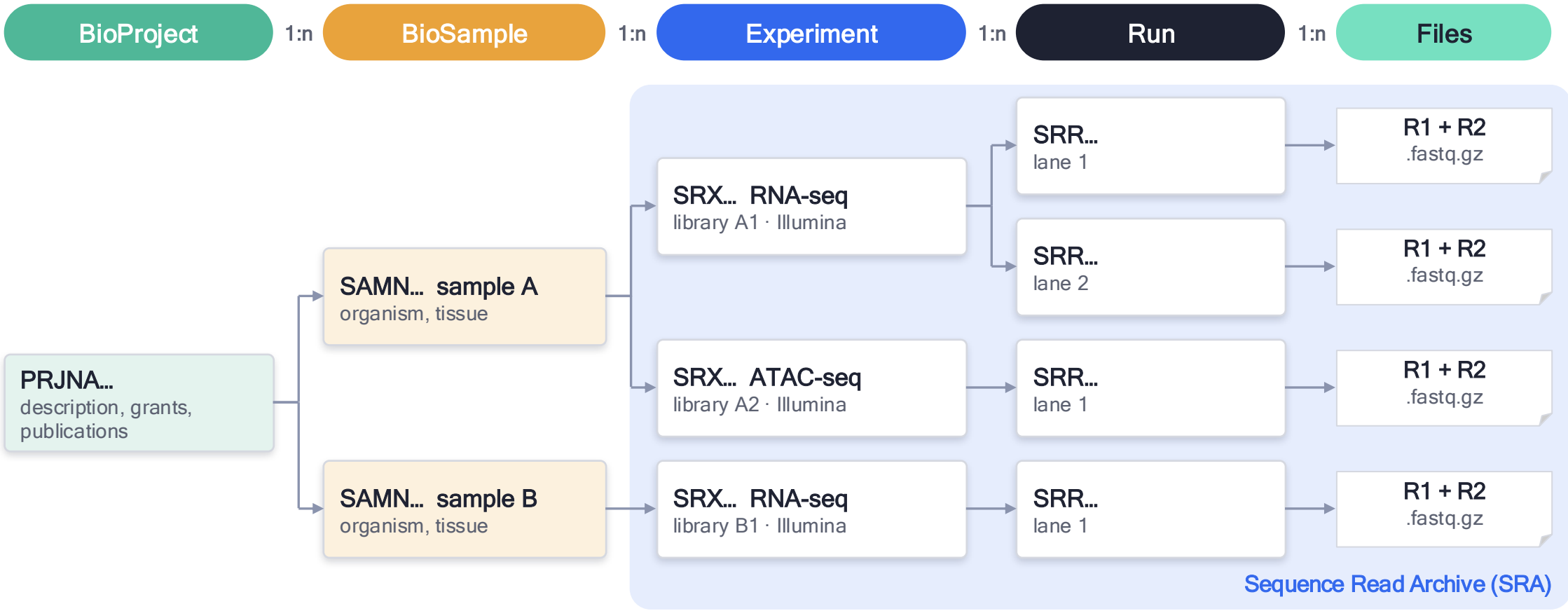
GenBank

NCBI, Bethesda

Raw reads: **SRA**

Synchronised daily: submit to any partner → same record, same accession at all three

How SRA organises data



Each run → **exactly 1** experiment → **exactly 1** BioSample

Several runs of one experiment = technical replicates → merge FASTQs

Accession numbers decoded

Level	NCBI	ENA	DDBJ	Describes
BioProject	PRJNA	PRJEB	PRJDB	the whole research project
BioSample	SAMN	SAMEA	SAMD	one biological sample
Study	SRP	ERP	DRP	the submitted study
Experiment	SRX	ERX	DRX	library + sequencing platform
Run	SRR	ERR	DRR	the actual data files (FASTQ)

One project → many samples → many experiments → one or more runs each

GEO own IDs

GSE series (experiment)

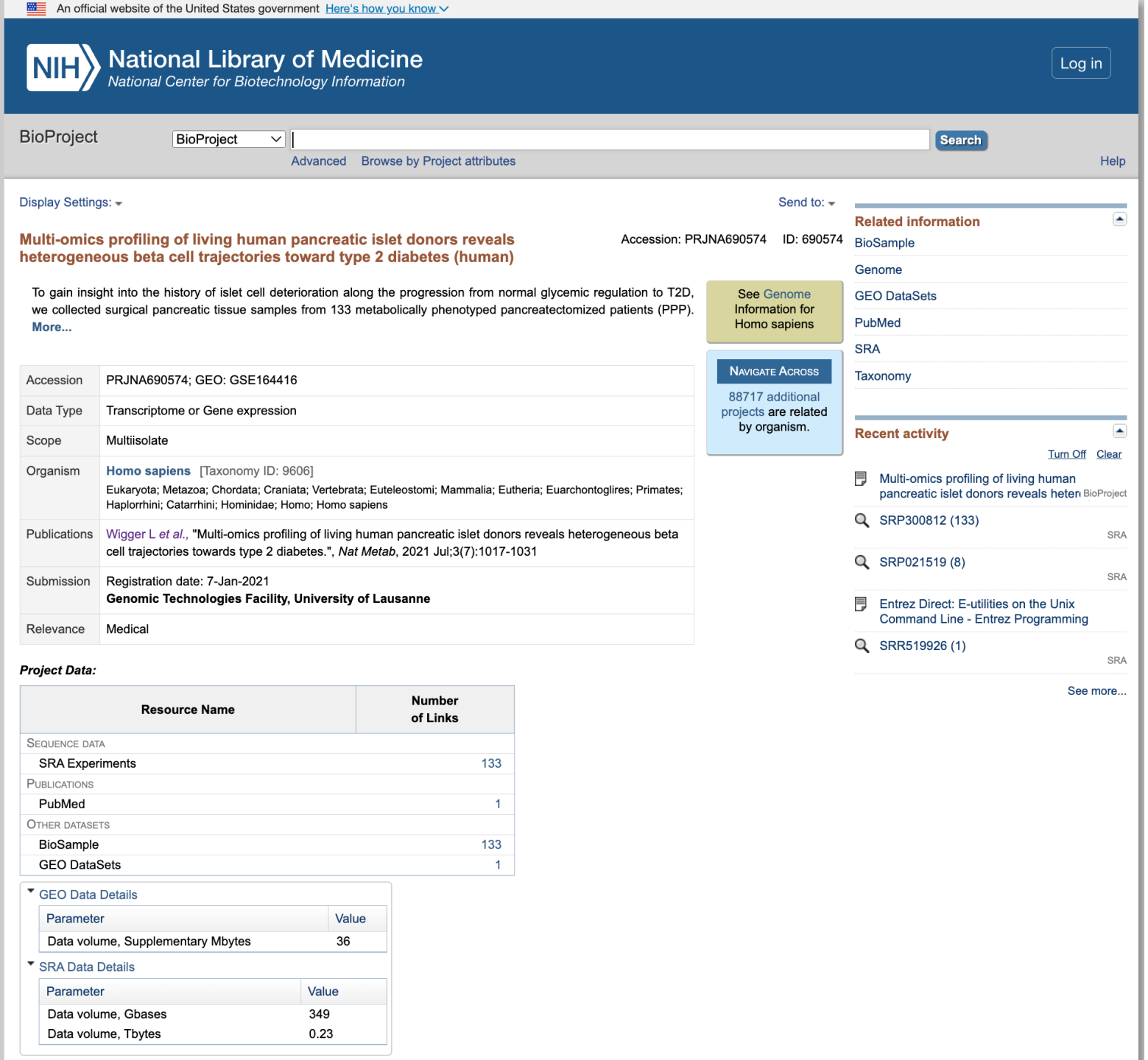
GSM sample

GPL platform

GSE ↔ PRJNA ↔ SRR are cross-linked: GEO holds processed data, SRA the reads

Tip: the first letter tells you the archive – S = NCBI, E = EBI, D = DDBJ

Screenshot of Bioproject



An official website of the United States government Here's how you know

National Library of Medicine
National Center for Biotechnology Information

Log in

BioProject

BioProject Search

Advanced Browse by Project attributes Help

Display Settings: Send to:

Multi-omics profiling of living human pancreatic islet donors reveals heterogeneous beta cell trajectories toward type 2 diabetes (human)

Accession: PRJNA690574 ID: 690574

To gain insight into the history of islet cell deterioration along the progression from normal glycemic regulation to T2D, we collected surgical pancreatic tissue samples from 133 metabolically phenotyped pancreatectomized patients (PPP). More...

See Genome Information for Homo sapiens

NAVIGATE ACROSS

88717 additional projects are related by organism.

Accession	PRJNA690574; GEO: GSE164416
Data Type	Transcriptome or Gene expression
Scope	Multiisolate
Organism	**Homo sapiens** [Taxonomy ID: 9606] Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo; Homo sapiens
Publications	Wigger L *et al.*, "Multi-omics profiling of living human pancreatic islet donors reveals heterogeneous beta cell trajectories towards type 2 diabetes.", *Nat Metab*, 2021 Jul;3(7):1017-1031
Submission	Registration date: 7-Jan-2021 **Genomic Technologies Facility, University of Lausanne**
Relevance	Medical

Project Data:

Resource Name	Number of Links
SEQUENCE DATA	
SRA Experiments	133
PUBLICATIONS	
PubMed	1
OTHER DATASETS	
BioSample	133
GEO DataSets	1

GEO Data Details

Parameter	Value
Data volume, Supplementary Mbytes	36

SRA Data Details

Parameter	Value
Data volume, Gbases	349
Data volume, Tbytes	0.23

Related information

BioSample
Genome
GEO DataSets
PubMed
SRA
Taxonomy

Recent activity

Turn Off Clear

Multi-omics profiling of living human pancreatic islet donors reveals heter BioProject
SRP300812 (133) SRA
SRP021519 (8) SRA
Entrez Direct: E-utilities on the Unix Command Line - Entrez Programming
SRR519926 (1) SRA

See more...

Example description of a project

DDBJ • Services SuperComputer Statistics Activities About Us • Terms Contact

home > bioproject > PRJNA690574

identifier	PRJNA690574
type	bioproject
sameAs	**GEO** GSE164416
organism	Homo sapiens
title	Multi-omics profiling of living human pancreatic islet donors reveals heterogeneous beta cell trajectories toward type 2 diabetes
description	To gain insight into the history of islet cell deterioration along the progression from normal glycemic regulation to T2D, we collected surgical pancreatic tissue samples from 133 metabolically phenotyped pancreatectomized patients (PPP). Gene expression profiles of islets isolated by laser capture microdissection (LCM) from resected and snap-frozen pancreas samples were assessed by RNA sequencing.Overall design: This study includes RNA-Seq samples from pancreatic islets of 133 human donors, stratified into four groups based on their diabetes status: 18 were non-diabetic (ND), 41 had impaired glucose tolerance (IGT), 35 had Type 3c diabetes (T3cD), and 39 had Type 2 diabetes (T2D). The group assignments are based on thresholds defined in the guidelines of the American Diabetes Association.For data analysis, a subset of 92 pancreatic islet samples was defined, which included only those samples in which the gene INS showed the highest expression (i.e., highest normalized counts value). Statistical analyses were performed both on the complete transcriptomics data set and on this restricted data set.
data type	Transcriptome or Gene expression
organization	
publication	34183850
external link	

Database landscape: where to find what

1 Raw reads

SRA · ENA · DRA

FASTQ / BAM / CRAM from sequencing projects, with run-level metadata

2 Processed data

GEO · ArrayExpress (BioStudies)

Count matrices, normalised data and sample annotations; link back to raw reads

3 Reference genomes

Ensembl · GENCODE · RefSeq · UCSC

Genome FASTA + GTF/GFF annotation – take both from the same release

4 Proteins

UniProtKB (Swiss-Prot / TrEMBL)

Curated protein sequences and function; Swiss-Prot is co-developed by SIB

5 Single-cell atlases

CZ CELLxGENE · Human Cell Atlas

Annotated, ready-to-use single-cell datasets for exploration and reuse

6 Controlled human data

EGA · dbGaP

Individual-level genomic and phenotypic data behind a Data Access Committee

Download raw reads: SRA-tools

SRA-tools *NCBI · works with any SRR / ERR / DRR accession*

command

```
prefetch SRR1234567                                  # 1. download the compressed .sra archive
fasterq-dump --split-files -e 4 -O fastq/ SRR1234567 # 2. convert to FASTQ, 4 threads
gzip fastq/*.fastq                                   # 3. output is uncompressed: compress it
```

output (illustrative)

```
spots read      : 21,438,077
reads read      : 42,876,154
reads written   : 42,876,154
$ ls fastq/
SRR1234567_1.fastq.gz   SRR1234567_2.fastq.gz
$ zcat fastq/SRR1234567_1.fastq.gz | head -4
@SRR1234567.1 1 length=101
NTGACCAGGCTTCAAGTTCCTGGAGCACCTGCGCAAGGACGTGATCCACCAGCACAGCGTCTC…
+SRR1234567.1 1 length=101
#FFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFF:FFFF,FFFFFFFFFFFF:FFFFFFFFF…
```

- **Spot** = one cluster; paired-end gives 2 reads per spot → files _1 and _2
- **Read names** are rewritten by SRA as @accession.spot (original Illumina IDs are lost)

Entrez Direct: a first example

Download a complete genome from its accession: the SARS-CoV-2 reference sequence, **NC_045512**

command

```
esearch -db nuccore -query 'NC_045512' \
  | efetch -format fasta > sars-cov-2_Wuhan-Hu-1.fasta
```

-db nuccore
search the nucleotide sequence database

-query 'NC_045512'
what to look for: here an accession

|
pass the result to the next command

-format fasta
output format (gb = annotated GenBank)

> file.fasta
save into a file instead of the screen

output

```
$ head -2 sars-cov-2_Wuhan-Hu-1.fasta
>NC_045512.2 Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1, complete genome
ATTAAAGGTTTATACCTTCCCAGGTAACAAACCAACCAACTTTCGATCTCTTGTAGATCTGTTCTCTAAA
$ grep -v '>' sars-cov-2_Wuhan-Hu-1.fasta | tr -d '\n' | wc -c      # genome length
29903
```

Shortcut if you already know the accession: `efetch -db nuccore -id NC_045512 -format fasta`

Entrez Direct: what goes through the pipe?

The pipe `|` sends the output of the command on its left as input to the command on its right. Let's run the two halves separately.

1 `esearch` searches the database

```
$ esearch -db sra -query PRJNA690574
```

Finds the 133 matching records, but keeps them on NCBI's server. It only prints a small "ticket" pointing to that result list.

output of step 1

```
<ENTREZ_DIRECT>
  <Db>sra</Db>                          ← which database
  <WebEnv>MCID_…</WebEnv>               ← your session on NCBI
  <QueryKey>1</QueryKey>                ← which result list
  <Count>133</Count>                    ← how many records
  <Step>1</Step>
</ENTREZ_DIRECT>
```

`|` passes this ticket, not the data: the next command uses it to ask NCBI for the 133 records

2 `efetch` downloads the records

```
$ … | efetch -format runinfo > runinfo.csv
```

`-format runinfo` = one CSV line per run; **`> file`** saves it

output of step 2

```
$ wc -l runinfo.csv
134 runinfo.csv                 # 1 header line + 133 runs
$ cut -d, -f1 runinfo.csv | head -3     # first column = Run
Run
SRR13380420
SRR13380421
```

For sensitive human data



EUROPEAN
GENOME-PHENOME
ARCHIVE

See also: https://ega-archive.org/about/projects-and-funders/federated-ega/

EGA

European Genome-phenome Archive

Run by EMBL-EBI and CRG (Barcelona)

Controlled access to human data with privacy concerns

US counterpart: **dbGaP**

Ideal for

- **Clinical studies**
- **Patient phenotypes**
- **Genomic variants & raw reads**

How to get access

1. **Find the dataset**
   search ega-archive.org for a study (EGAS) and dataset (EGAD)
2. **Request access**
   apply to the study's Data Access Committee (DAC)
3. **DAC approval**
   project reviewed; data access agreement signed
4. **Download**
   encrypted files (EGAF) with the pyEGA3 client

Quiz: 18 - 20

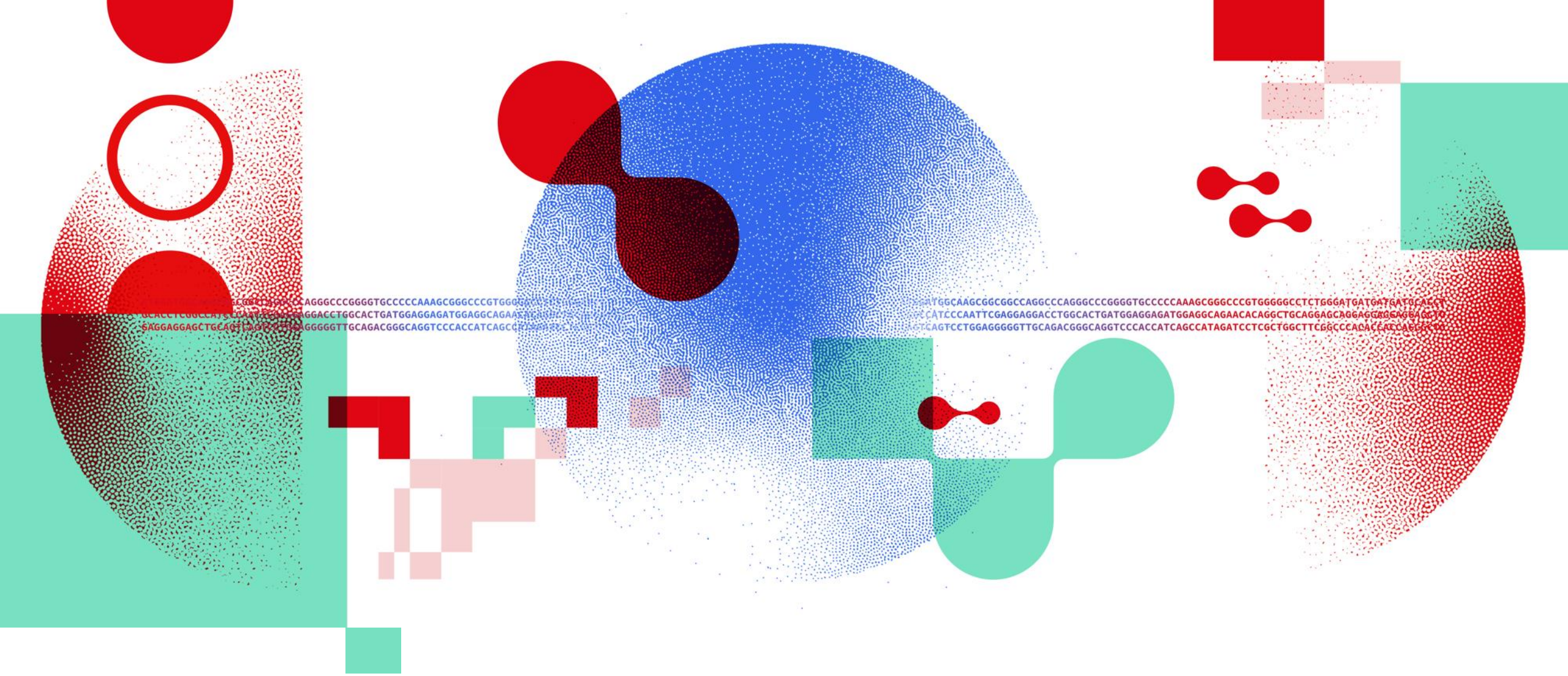
Which database is part of INSDC?

. . .

What command-line tool is used to download raw data from SRA?

Summary

QC	ensures reliable sequencing data: quality, length, adapters, overrepresented sequences
Trimming	removes low-quality bases, adapters and poly-G – fastp (or cutadapt, Trimmomatic …)
INSDC	DDBJ · ENA · GenBank exchange data daily; raw reads live in DRA / ENA / SRA
EGA	controlled access to sensitive human data through a Data Access Committee
CLI tools	SRA-tools, ENA links / fetchngs and Entrez Direct fetch data reproducibly



Thank you

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